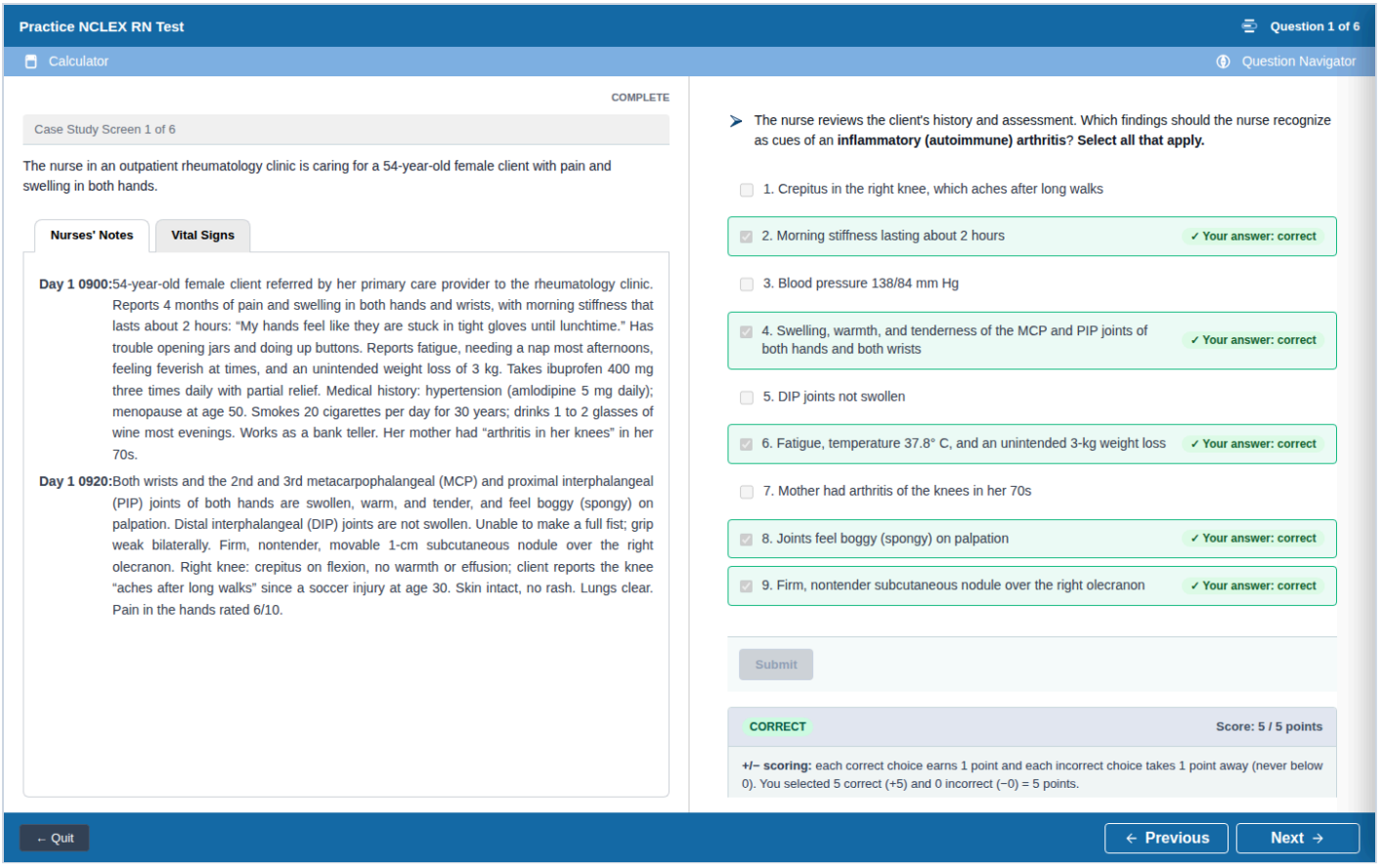
NURS 1017 Unit 6 Case Study 1: Rheumatoid arthritis (review copy)

For each screen: the question with the answer key filled in and submitted, then the full rationale as students see it after Submit. All 6 screens score full marks with their answer keys and the case passes the readiness check.

Screen 2 is the matrix multiple-response question comparing rheumatoid arthritis, osteoarthritis and osteomyelitis.

Screen 1 of 6 (Recognize cues)



Screen 1: rationale

CORRECT

Score: 5 / 5 points

+/− scoring: each correct choice earns 1 point and each incorrect choice takes 1 point away (never below 0). You selected 5 correct (+5) and 0 incorrect (−0) = 5 points.

Rationale:

Cues of inflammatory (autoimmune) arthritis:

Prolonged morning stiffness: stiffness lasting more than 1 hour after rest is typical of inflammatory arthritis. Fluid and inflammatory mediators build up in the inflamed synovium overnight. In osteoarthritis, stiffness usually lasts less than 30 minutes.

Symmetric small-joint swelling: rheumatoid arthritis (RA) classically involves the same small joints on both sides (MCP, PIP, wrists). Warmth and tenderness show active synovitis.

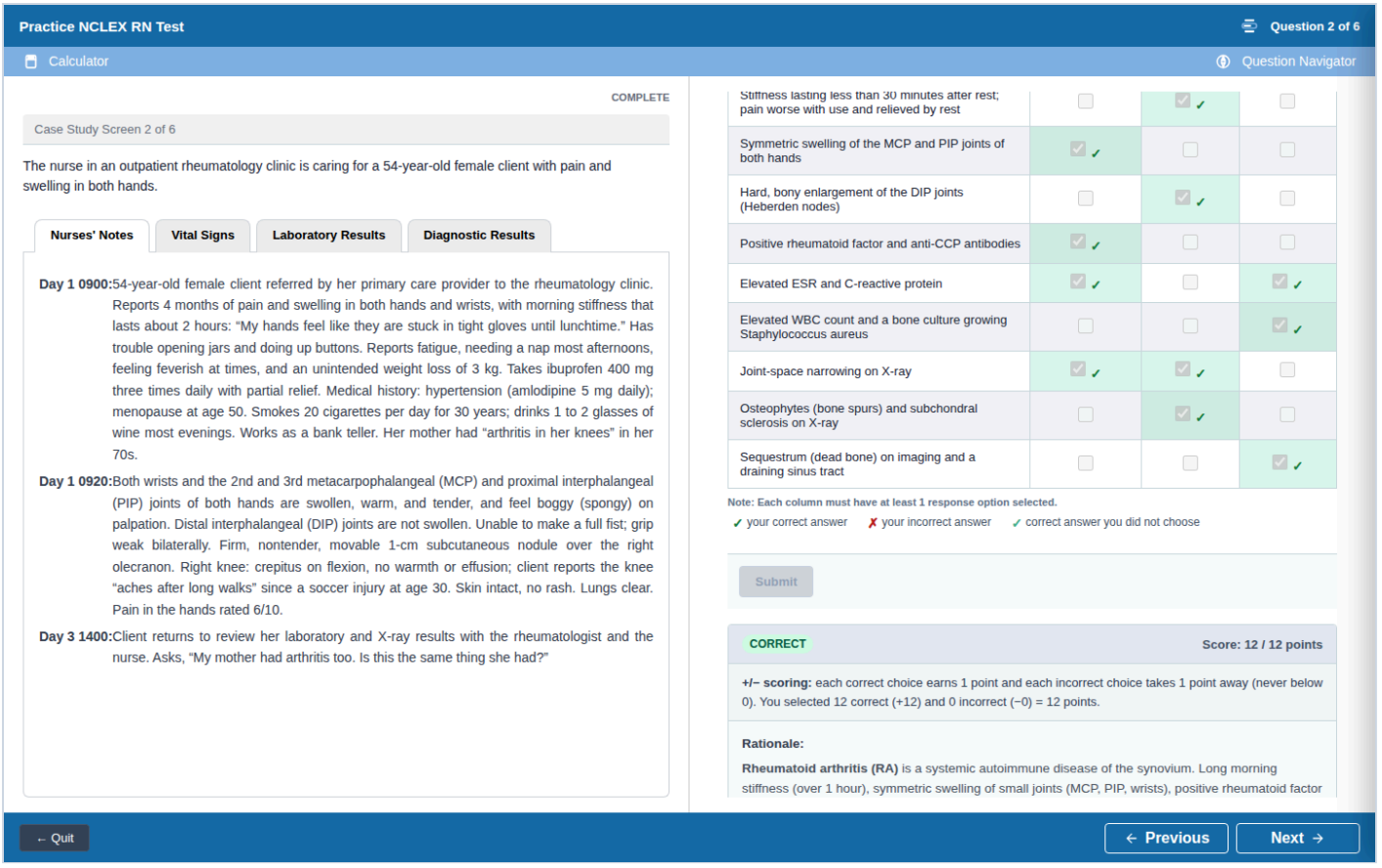
Boggy joints: a soft, spongy feel comes from thickened, inflamed synovium and joint fluid. Osteoarthritis joints feel hard and bony.

Systemic signs: fatigue, low-grade fever and weight loss reflect circulating inflammatory cytokines (TNF-α, IL-6). RA is a systemic disease, not only a joint disease.

Rheumatoid nodule: a firm, nontender nodule over a pressure point such as the olecranon is an extra-articular feature of RA, seen mostly in seropositive disease.

Not cues: crepitus and pain after activity in one weight-bearing joint (the right knee) after an old injury point to osteoarthritis. The mother's knee arthritis in her 70s also sounds like osteoarthritis. Sparing of the DIP joints is expected in RA and is not a sign of disease. The blood pressure reflects her treated hypertension.

Screen 2 of 6 (Analyze cues)



Screen 2: rationale

CORRECT

Score: 12 / 12 points

+/− scoring: each correct choice earns 1 point and each incorrect choice takes 1 point away (never below 0). You selected 12 correct (+12) and 0 incorrect (−0) = 12 points.

Rationale:

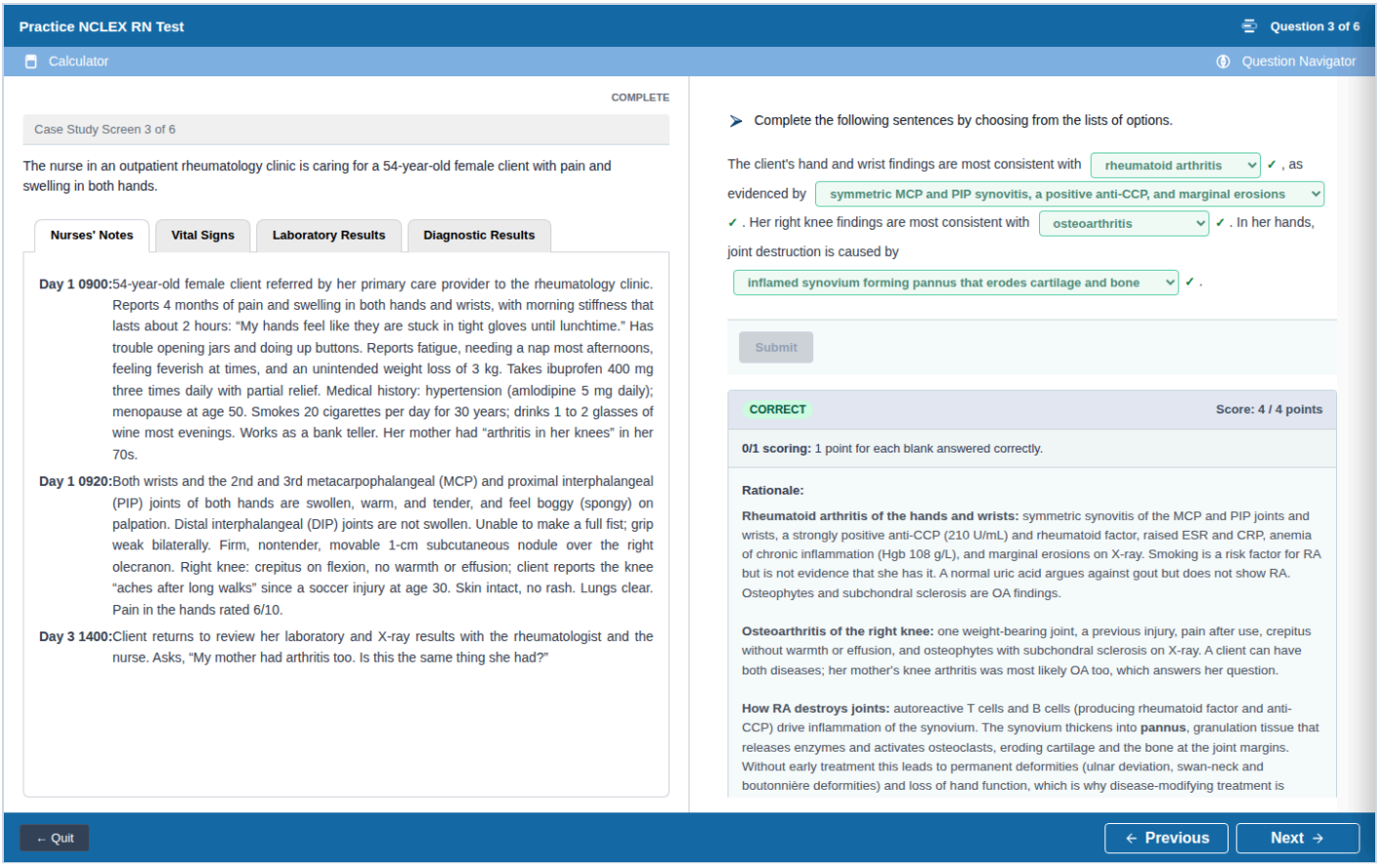
Rheumatoid arthritis (RA) is a systemic autoimmune disease of the synovium. Long morning stiffness (over 1 hour), symmetric swelling of small joints (MCP, PIP, wrists), positive rheumatoid factor and especially **anti-CCP antibodies** (highly specific for RA) are typical. Inflammation raises the **ESR and CRP**. X-rays show periarticular osteopenia, uniform **joint-space narrowing** and marginal erosions (as in this client's hands).

Osteoarthritis (OA) is a degenerative disease of articular cartilage, not an autoimmune one. Stiffness after rest is brief (under 30 minutes), and pain worsens with use and eases with rest. Bony enlargement of the DIP joints (**Heberden nodes**) and PIP joints (Bouchard nodes) is typical. ESR and CRP are usually normal and serology is negative. X-rays show uneven **joint-space narrowing**, **osteophytes** and subchondral sclerosis (as in this client's right knee).

Osteomyelitis is an infection of bone, most often by *Staphylococcus aureus*, spread through the blood or from an open fracture, surgery or a nearby wound. Infection causes fever, an elevated **WBC count**, and raised **ESR and CRP** (used to follow the response to antibiotics). A bone biopsy and culture identify the organism. In chronic osteomyelitis, dead bone (**sequestrum**) is walled off by new bone (involucrum), and pus may drain through a **sinus tract** to the skin.

Shared findings: ESR and CRP rise with inflammation from any cause (RA and osteomyelitis). Joint-space narrowing occurs in both RA and OA, because both destroy articular cartilage, by different mechanisms.

Screen 3 of 6 (Prioritize hypotheses)



Screen 3: rationale

CORRECT

Score: 4 / 4 points

0/1 scoring: 1 point for each blank answered correctly.

Rationale:

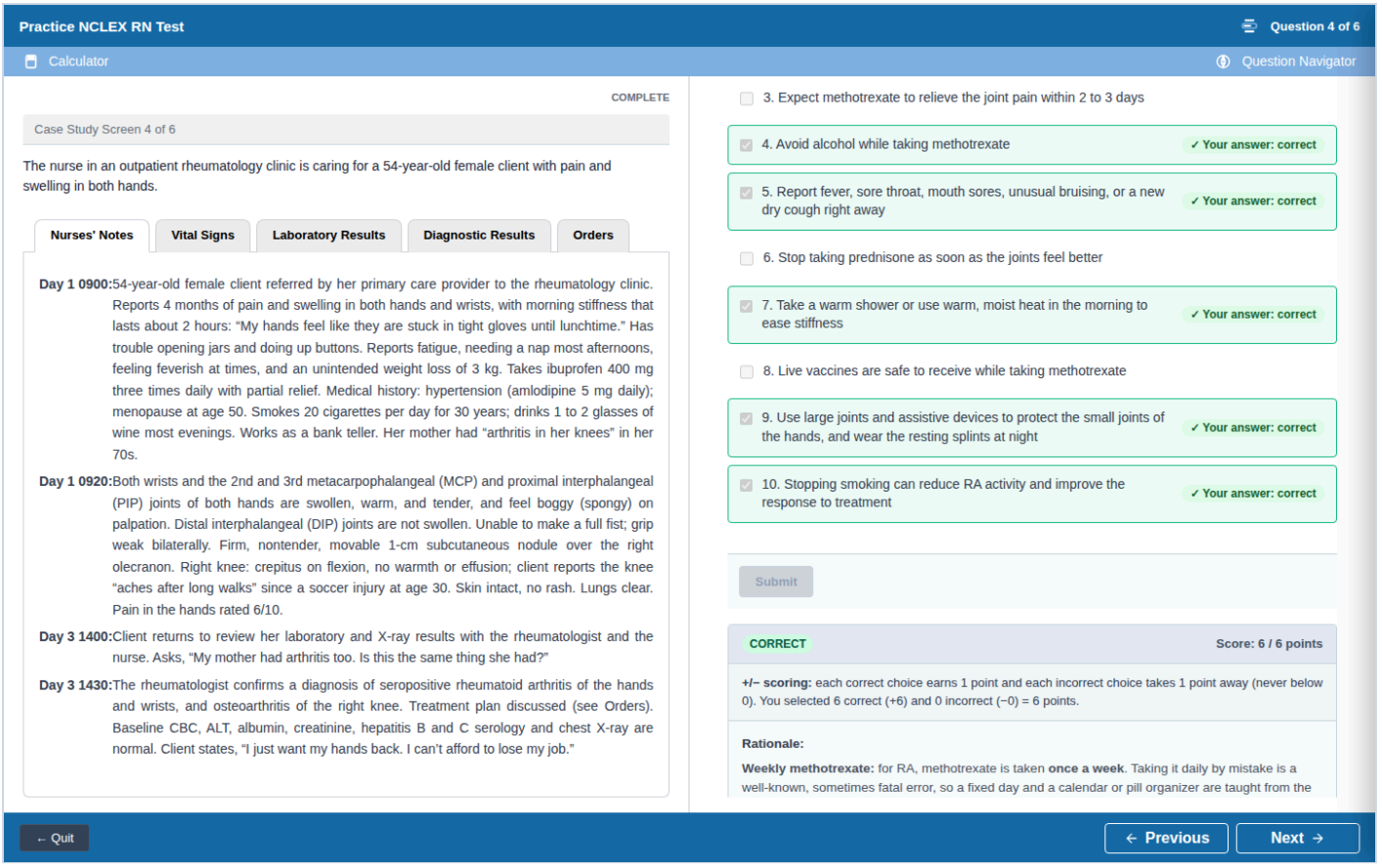
Rheumatoid arthritis of the hands and wrists: symmetric synovitis of the MCP and PIP joints and wrists, a strongly positive anti-CCP (210 U/mL) and rheumatoid factor, raised ESR and CRP, anemia of chronic inflammation (Hgb 108 g/L), and marginal erosions on X-ray. Smoking is a risk factor for RA but is not evidence that she has it. A normal uric acid argues against gout but does not show RA. Osteophytes and subchondral sclerosis are OA findings.

Osteoarthritis of the right knee: one weight-bearing joint, a previous injury, pain after use, crepitus without warmth or effusion, and osteophytes with subchondral sclerosis on X-ray. A client can have both diseases; her mother's knee arthritis was most likely OA too, which answers her question.

How RA destroys joints: autoreactive T cells and B cells (producing rheumatoid factor and anti-CCP) drive inflammation of the synovium. The synovium thickens into **pannus**, granulation tissue that releases enzymes and activates osteoclasts, eroding cartilage and the bone at the joint margins. Without early treatment this leads to permanent deformities (ulnar deviation, swan-neck and boutonnière deformities) and loss of hand function, which is why disease-modifying treatment is started promptly.

Why not the others: mechanical wear with bone spurs is OA; urate crystals cause gout; bacterial infection of bone is osteomyelitis.

Screen 4 of 6 (Generate solutions)



Screen 4: rationale

CORRECT

Score: 6 / 6 points

+/− scoring: each correct choice earns 1 point and each incorrect choice takes 1 point away (never below 0). You selected 6 correct (+6) and 0 incorrect (−0) = 6 points.

Rationale:

Weekly methotrexate: for RA, methotrexate is taken **once a week**. Taking it daily by mistake is a well-known, sometimes fatal error, so a fixed day and a calendar or pill organizer are taught from the start. Folic acid reduces mouth sores, nausea and liver enzyme rises.

Alcohol: methotrexate and alcohol are both toxic to the liver, so alcohol should be avoided (the client drinks most evenings).

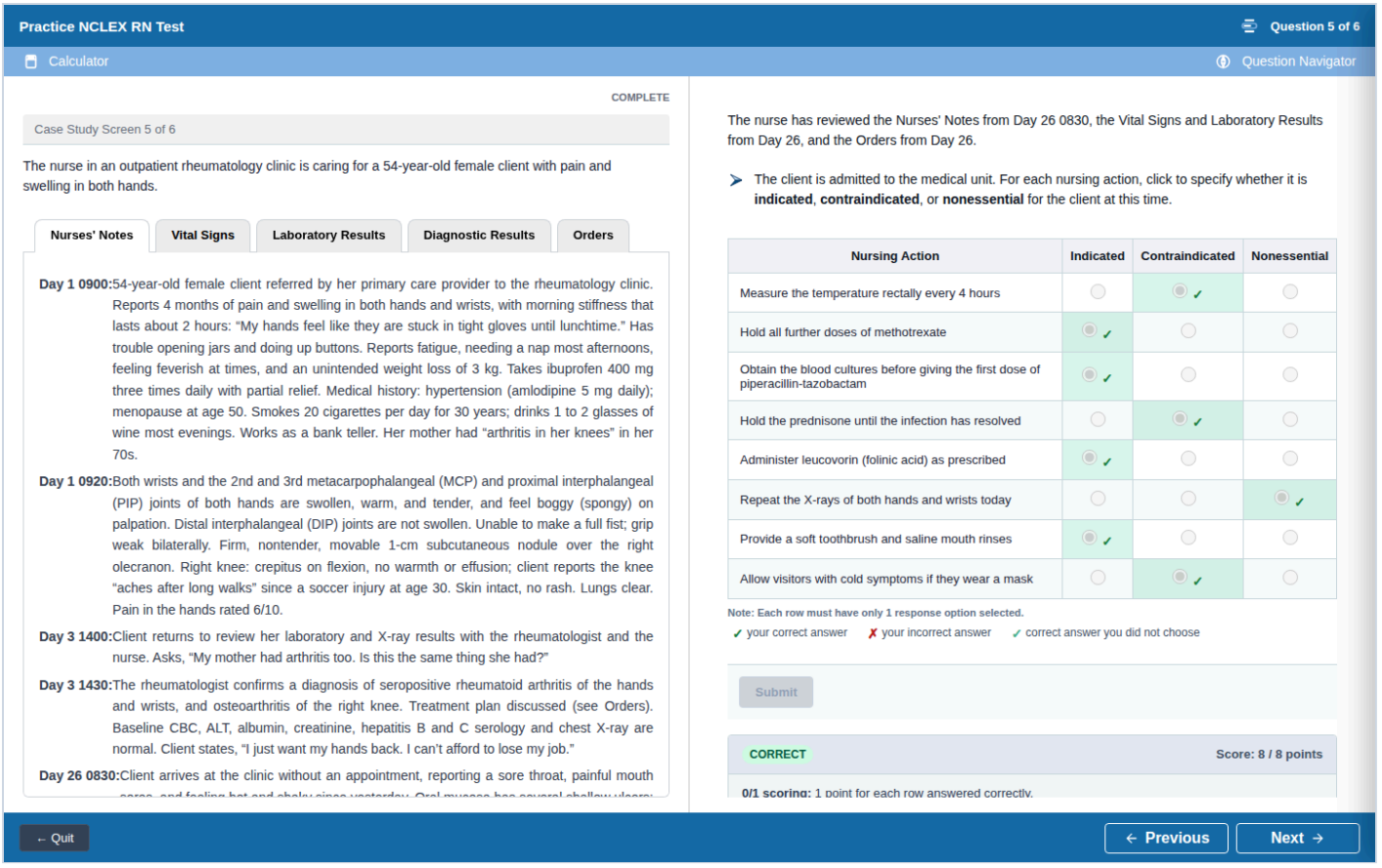
Warning signs: methotrexate can suppress the bone marrow (infection, bleeding), cause mucositis, and rarely cause pneumonitis (dry cough, shortness of breath). These are reported promptly.

Heat and joint protection: warm, moist heat eases morning stiffness. Using larger joints (carrying a bag on the forearm), jar openers and built-up handles, and resting splints at night protect inflamed joints and keep them in a functional position. Balancing rest with gentle range-of-motion exercise maintains function.

Smoking: smoking increases RA activity and the production of anti-CCP antibodies, and lowers the response to methotrexate. It is also a risk factor for osteoporosis.

Incorrect: resting joints flexed on pillows leads to flexion contractures. Methotrexate takes about 6 to 12 weeks to work, which is why prednisone is used as a short bridge. Prednisone must be tapered as prescribed, never stopped suddenly (adrenal insufficiency). Live vaccines are avoided during immunosuppressive therapy; inactivated and recombinant vaccines (influenza, pneumococcal, recombinant zoster) are given.

Screen 5 of 6 (Take action)



Screen 5: rationale

CORRECT

Score: 8 / 8 points

0/1 scoring: 1 point for each row answered correctly.

Rationale:

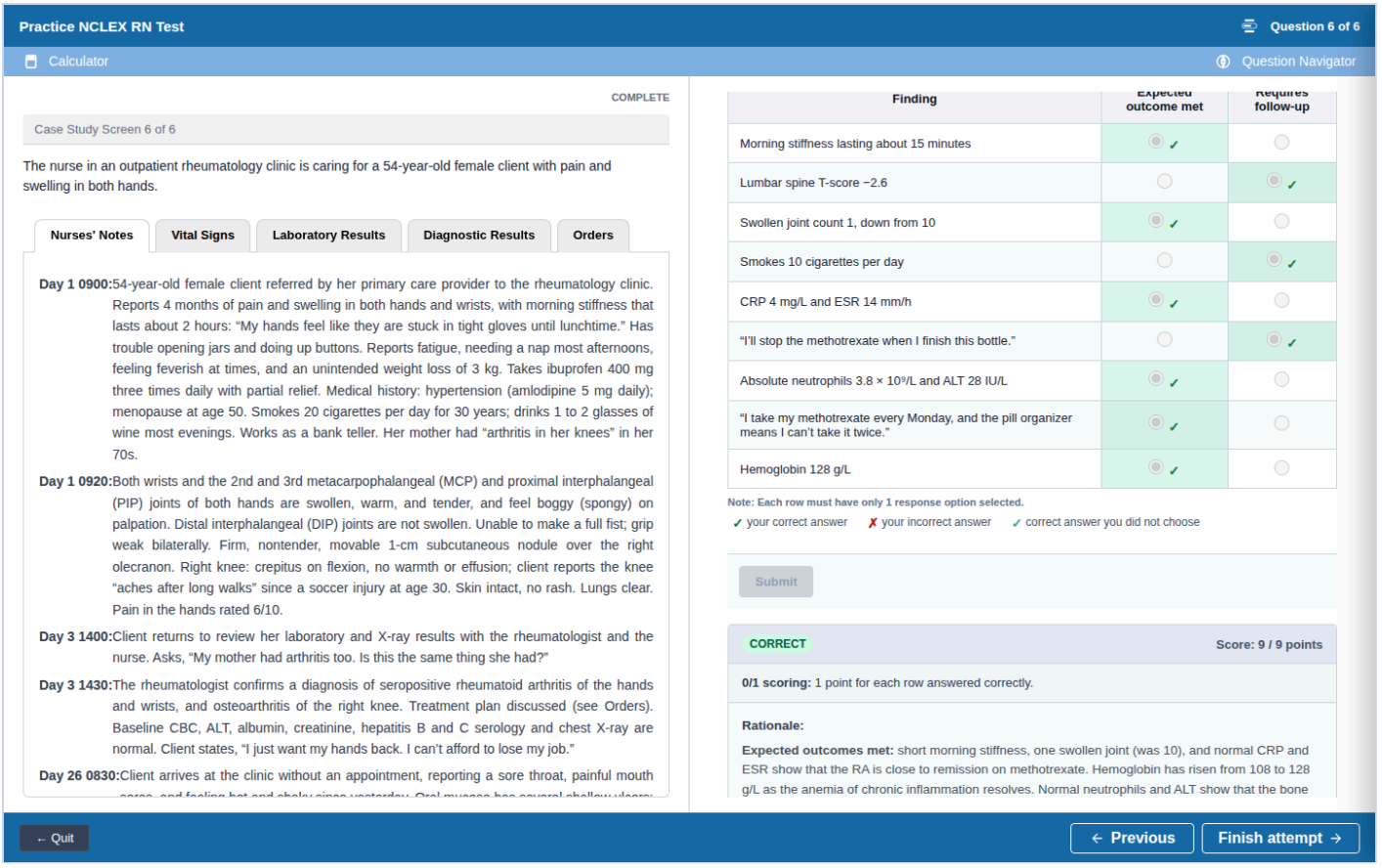
The client took methotrexate daily instead of weekly for 5 days and now has **methotrexate toxicity**: severe neutropenia (absolute neutrophils 0.4 × 10^9/L), thrombocytopenia (74 × 10^9/L), mucositis, raised ALT, and fever of 38.6° C with tachycardia (**febrile neutropenia**, a medical emergency).

Indicated: stop the drug (hold methotrexate). Give **leucovorin**, the reduced folate that bypasses the enzyme methotrexate blocks, as the antidote. Draw **blood cultures before the first antibiotic dose** so the organism can be identified, then start the antibiotic without delay. A soft toothbrush and saline rinses reduce mucosal injury, bleeding and infection.

Contraindicated: rectal temperatures can tear the mucosa and introduce bacteria in a neutropenic client with low platelets. Stopping long-term prednisone suddenly can cause adrenal crisis, and illness raises steroid needs; the order is to continue it. Visitors with respiratory infections should not visit a severely neutropenic client, even with a mask.

Nonessential: repeat hand X-rays do not change the management of this emergency; joint damage is followed over months.

Screen 6 of 6 (Evaluate outcomes)



Screen 6: rationale

CORRECT

Score: 9 / 9 points

0/1 scoring: 1 point for each row answered correctly.

Rationale:

Expected outcomes met: short morning stiffness, one swollen joint (was 10), and normal CRP and ESR show that the RA is close to remission on methotrexate. Hemoglobin has risen from 108 to 128 g/L as the anemia of chronic inflammation resolves. Normal neutrophils and ALT show that the bone marrow and liver have recovered and that weekly methotrexate is tolerated. Her statement about taking it every Monday with a pill organizer shows she understands the weekly schedule.

Requires follow-up:

Osteoporosis: a T-score of −2.5 or lower is osteoporosis. RA itself, glucocorticoid use, postmenopausal status, smoking and alcohol all reduce bone density. The nurse reports the result so the provider can consider bone-protecting medication (for example, a bisphosphonate), and reinforces calcium, vitamin D, weight-bearing exercise and fall prevention.

Smoking: she has halved her smoking, which is progress, but continued smoking worsens RA and bone loss. Ongoing cessation support is needed.

Plan to stop methotrexate: RA is a chronic disease. Methotrexate controls it but does not cure it; stopping it on her own would likely cause a flare and further joint damage. Any change must be decided with the rheumatologist.